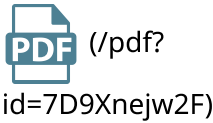


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Learning Direction-Specific Cross-Modal Interactions for Vision-Language Tasks



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19 Jun 2026 (modified: 25 Jun 2026) WACV 2027 Conference Round 1 Submission
 Conference Round 1, Senior Area Chairs, Area Chairs, Reviewers, Authors
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Subject Area: Vision, Language, and Multimodal AI
Student Paper: No
Track: Algorithms Track

Abstract:
Multimodal representation learning requires models to fuse heterogeneous information sources, such as images and text. However, many existing fusion methods use symmetric cross-modal interaction mechanisms that treat image-to-text and text-to-image interactions as equivalent, even though these directions can serve different functional roles. In this paper, we propose an asymmetric cross-modal learning framework that uses independent cross-attention modules for each direction of multimodal exchange, enabling direction-specific representations for multimodal reasoning. We evaluate the framework on CLEVR and Flickr30k through controlled comparisons against a symmetric shared-attention baseline, including frozen-encoder experiments, multi-seed evaluation, modality-blind ablations, question-type analysis, and image--text retrieval evaluation. Results show strong gains on CLEVR, especially for frozen-encoder compositional reasoning, indicating that asymmetric fusion is effective when language defines the reasoning task and vision provides the evidence. In contrast, Flickr30k retrieval results show that symmetric fusion remains competitive for more balanced image--text matching. These findings suggest that direction-specific cross-modal attention is most effective when modalities play distinct functional roles.

Submission Number: 1884

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Meta Review of Submission1884 by Area Chair vrwc

Meta Review by Area Chair vrwc 📅 02 Aug 2026, 14:38 (modified: 09 Aug 2026, 22:05)

👁 Senior Area Chairs, Area Chairs, Program Chairs, Authors, Reviewers Submitted

📄 Revisions (/revisions?id=CYGRjkO8Yh)

Metareview:

This paper receives two borderline rejects and two weak rejects. The current experiments do not conclusively demonstrate that direction-specific fusion itself is responsible for the improvements. The requested revisions are primarily aimed at: add a capacity-matched symmetric baseline to isolate the effect of directional fusion from increased model capacity, clarify the Flickr30k retrieval protocols and explain the conflicting results, improve experimental reporting (splits, seeds, checkpoints, table provenance), add stronger evaluation on real-world VQA datasets (e.g., complete VQA 2.0 results), re-consider the generality especially given the mixed Flickr30k results, etc.

Final Recommendation: Recommend resubmission in round 2 with updated paper and rebuttal

a simple cross attention method

Official Review by Reviewer XZ7H 📅 30 Jul 2026, 23:59 (modified: 09 Aug 2026, 21:48)

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Reviewer XZ7H, Authors

📄 Revisions (/revisions?id=umoJBGHMmn)

Paper Summary:

The paper studies a clear and potentially useful question: whether image-to-text and text-to-image cross-modal interactions should share the same attention parameters. The proposed asymmetric design is simple and easy to understand, and the large gains on CLEVR, especially in the frozen-encoder setting, suggest that direction-specific fusion may benefit tasks where language specifies the reasoning objective and vision provides the evidence. However, the current experiments do not convincingly isolate the effect of asymmetry itself, because the asymmetric model uses a larger fusion parameter budget than the symmetric baseline, and no parameter-matched comparison is provided. The retrieval results are also protocol-dependent, with the symmetric model outperforming the asymmetric one under the more faithful pairwise scoring setup. In addition, inconsistencies between the reported CLEVR results and the incomplete VQA evaluation reduce confidence in the empirical conclusions. Overall, the paper is clearly written and explores an interesting architectural hypothesis, but stronger capacity-controlled baselines, clearer experimental reporting, and comparisons with standard multimodal fusion methods are necessary before the main claim can be fully supported.

Paper Strengths:

he paper addresses a clear and relevant question about whether image-to-text and text-to-image interactions should share parameters. The asymmetric design is simple, intuitive, and easy to implement, while the strong CLEVR gains and additional multi-seed, modality-blind, question-type, and retrieval analyses provide useful evidence that direction-specific fusion can help when the two modalities play different functional roles

Major Weaknesses:

The main limitation is that the asymmetric model has more parameters than the symmetric baseline, so the experiments do not clearly separate the benefit of direction-specific modeling from increased capacity.

Minor Weaknesses:

The novelty is relatively limited, several experimental results are inconsistent or incomplete, and the asymmetric model performs worse under the more appropriate pairwise Flickr30k retrieval protocol, weakening the generality of the main claim.

Preliminary Recommendation: 2: Weak Reject

Preliminary Justification:

see weakness

Resubmission: Yes, the paper is of reasonable quality but requires some revisions before it can be accepted.

Revisions For Resubmission:

Authors should consider the weakness

Confidence Level: 4 - High Confidence: The reviewer has strong expertise in the area. They are highly familiar with the relevant literature and can critically evaluate the paper.

Clear Insights on CLEVR but Limited by Capacity Matching and Benchmark Scope

Official Review by Reviewer Ziyi 📅 28 Jul 2026, 14:10 (modified: 09 Aug 2026, 21:48)

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Reviewer Ziyi, Authors

📄 Revisions (/revisions?id=GrY81WJIsf)

Paper Summary:

This paper investigates the functional asymmetry of cross-modal interactions in vision-language tasks. The authors argue that standard symmetric cross-attention mechanisms, which share parameters between image-to-text and text-to-image directions, ignore the distinct roles played by each modality. To address this, they propose an asymmetric fusion framework employing independent, unshared cross-attention modules for each interaction direction.

The approach is evaluated on CLEVR (compositional VQA) and Flickr30k (image-text retrieval) across frozen and unfrozen encoder settings. The paper incorporates multi-seed evaluations, modality-blind ablations, question-type breakdowns, and retrieval protocol analyses. Results show substantial performance gains on CLEVR in the frozen-encoder regime (improving accuracy from 60.83% to 87.85%), whereas symmetric fusion remains superior under standard pairwise retrieval on Flickr30k.

Paper Strengths:

1. Clear Conceptual Motivation: The paper provides a well-reasoned intuition for directional cross-attention, distinguishing between context-guided visual localization ($T \rightarrow I$) and evidence-guided semantic interpretation ($I \rightarrow T$).
2. Significant Gains on Compositional Reasoning: The asymmetric model achieves a notable +27.02 improvement over the symmetric baseline on CLEVR with frozen encoders, highlighting potential optimization bottlenecks in shared parameter designs for VQA.
3. Thorough Empirical Diagnostics: The inclusion of multi-seed stability tests, detailed question-type breakdowns, and modality-blind ablations provides valuable diagnostic insights into how the model utilizes visual and textual inputs.
4. Discussion: The authors transparently report scenarios where asymmetric fusion underperforms (such as Flickr30k pairwise retrieval) and avoid overclaiming the universal superiority of their method.

Major Weaknesses:

1. Confounded Evaluation Due to Unmatched Parameter Budgets: By unsharing parameters across the two cross-attention directions, the proposed asymmetric module effectively doubles the parameter budget of the fusion layer relative to the symmetric baseline. While the authors acknowledge this capacity difference in Section 6, the manuscript lacks a capacity-matched baseline (for example, a symmetric model with two stacked cross-attention layers or doubled hidden dimensions). Consequently, it is difficult to isolate whether the observed gains on CLEVR stem strictly from directional specialization or simply from increased model capacity.
2. Discrepancy Between Dataset Description and Main Experimental Scope: Section 4.2 provides a detailed description of the real-world VQA 2.0 dataset. However, VQA 2.0 is notably absent from the primary validation and multi-seed result tables (Tables 2 and 3), which rely exclusively on the synthetic CLEVR benchmark. VQA 2.0 appears only as a single-seed ablation snippet in Table 5. Demonstrating effectiveness primarily on synthetic data limits confidence in the generalizability of the findings to real-world visual-language reasoning.
3. Incremental Architectural Novelty: From an architectural standpoint, unsharing weights between two standard Transformer cross-attention blocks ($CrossAttn_{A \rightarrow B}$ and $CrossAttn_{B \rightarrow A}$) is a relatively simple modification. The primary value of the manuscript lies in its empirical analysis rather than novel algorithmic or architectural design.
4. Performance Degradation on Mutual Matching Tasks: On Flickr30k under the standard pairwise scoring protocol (which is the more faithful protocol for candidate-level fusion architectures), the asymmetric model exhibits a noticeable drop in performance across all recall metrics (for instance, R@1 drops from 42.6 to 34.1 for I2T). This

indicates that unsharing parameters may act as a negative regularizer when modalities play balanced, non-directional roles.

Minor Weaknesses:

1. The repository link at the end of Section 1 is not accessible.
2. Duplicate Citation: References [23] and [24] are duplicate entries referring to the exact same paper (LanguageBind).
3. In Section 5.3, there is a leftover word typo: "This Removing the image...".

Preliminary Recommendation: 3: Borderline Reject

Preliminary Justification:

While the paper presents a clear functional motivation and impressive empirical gains on the synthetic CLEVR dataset, the evaluation suffers from two main shortcomings: the lack of parameter-capacity-matched baselines and the reliance on synthetic data for the main benchmarks despite describing VQA 2.0 in detail.

To move this paper toward acceptance, the authors should address the following during the rebuttal:

1. Capacity-Matched Baselines: Provide CLEVR results for a capacity-matched symmetric baseline (e.g., a 2-layer symmetric cross-attention module or widened hidden dimension) to disentangle parameter capacity effects from directional specialization.
2. Real-World VQA Validation: Provide complete, multi-seed evaluation results on the full VQA 2.0 dataset to validate the approach beyond synthetic environments.

Resubmission: Yes, the paper is of reasonable quality but requires some revisions before it can be accepted.

Revisions For Resubmission:

Please refer to the Major Weakness section.

Confidence Level: 4 - High Confidence: The reviewer has strong expertise in the area. They are highly familiar with the relevant literature and can critically evaluate the paper.

Review

Official Review by Reviewer m3rX  22 Jul 2026, 22:05 (modified: 09 Aug 2026, 21:48)

 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Reviewer m3rX, Authors

 Revisions (/revisions?id=nNQgbfe3Z8)

Paper Summary:

This paper studies whether image-to-text and text-to-image interactions should use different cross-attention modules instead of sharing parameters. The authors propose an asymmetric fusion architecture and compare it with the conventional shared-attention design on CLEVR and Flickr30k. The results show that asymmetric fusion significantly improves compositional reasoning on CLEVR but provides mixed results on image-text retrieval.

Paper Strengths:

1. The paper studies a clear research question and keeps the experimental setting well controlled.
2. The authors honestly discuss both positive and negative results instead of claiming that the proposed method always performs better.

Major Weaknesses:

1. The setting of the paper seems a bit out-dated. With the MLLMs being the standard architecture, I am not sure if these architectures are still relevant. Additionally, the experimental setting, including both training and evaluations, are also a bit outdated and more recent settings like Llava's, Qwen's may be considered.
2. Related to the first point, the experiments are limited to relatively small benchmarks. It would be interesting to verify whether the conclusion still holds for modern large vision-language models.

Minor Weaknesses:

1. The architectural novelty is relatively limited since the main idea is simply to separate the cross-attention modules for the two directions.

Preliminary Recommendation: 2: Weak Reject

Preliminary Justification:

The paper conducts some well-controlled experimental studies, but the architectures are relatively old and it is unclear how the conclusions transfer to state-of-the-art models.

Resubmission: No, the required revisions are too extensive for a resubmission in the time that is available.

Revisions For Resubmission:

N/A

Confidence Level: 2 - Low Confidence: The reviewer has some familiarity with the topic but does not have deep expertise. They might understand the paper but feel unsure about their evaluation.

Controlled study of direction-specific cross-modal interactions, but the central attribution and retrieval evaluation need clarification

Official Review by Reviewer 9SKd 22 Jul 2026, 10:56 (modified: 09 Aug 2026, 21:48)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Reviewer 9SKd, Authors

Revisions (/revisions?id=zAHdHIqti4)

Paper Summary:

This paper studies whether the two opposite directions of vision-language interaction should share parameters. It proposes an asymmetric fusion framework that uses separate cross-attention modules for text-to-image and image-to-text interaction, and compares it with a symmetric baseline that reuses one cross-attention module. The experiments cover CLEVR visual question answering and Flickr30k retrieval, with frozen and fine-tuned encoders, multi-seed evaluation, modality-blind ablations, CLEVR question-type analysis, and two retrieval protocols. The asymmetric design substantially improves CLEVR performance, especially with frozen encoders, whereas the Flickr30k conclusion depends on the protocol: the symmetric model is stronger under pairwise scoring. The paper concludes that direction-specific fusion is most useful when language specifies the reasoning task and vision provides the evidence.

Paper Strengths:

1. The paper isolates a clear and practically relevant architectural question: whether opposite directions of cross-modal attention should share parameters. The proposed change is simple and easy to understand.
2. The CLEVR study is thoughtfully controlled. Frozen-encoder experiments, multiple seeds, modality-blind ablations, and question-type analysis provide consistent evidence that the asymmetric design can help language-guided compositional visual reasoning.
3. The paper reports unfavorable as well as favorable results. In particular, symmetric fusion outperforms asymmetric fusion under pairwise Flickr30k scoring, and the paper appropriately narrows its claim rather than asserting universal superiority.

Major Weaknesses:

1. The symmetric and asymmetric fusion modules have different parameter counts, so the large CLEVR gain cannot yet be attributed solely to direction-specific parameterization; it may partly reflect greater capacity. The limitation is acknowledged in Section 6, but it affects the central claim. A lightweight parameter-matched comparison would address this directly, for example by reducing the width of the asymmetric block or widening the shared block. Reporting a few seeds in the frozen CLEVR setting would be sufficient.
2. The definition of the “fusion-inclusive CLIP-style” Flickr30k protocol is not sufficiently clear. Since cross-attention outputs naturally depend on an image-text pair, the paper should specify how the image and text representations are constructed independently, whether their construction uses the ground-truth paired item, and how candidates are scored. A short formula or pseudocode plus one query example would make the protocol reproducible. The pairwise protocol appears more faithful to a fusion model and should be the primary basis for the retrieval conclusion. If the other protocol uses known pairing information, it should be explicitly described as a diagnostic rather than standard retrieval.
3. Several reported settings and numbers need clarification for reproducibility. For example, the frozen CLEVR symmetric result in Table 2 is (60.83 $\pm$ 0.82), whereas the comparable “Full input” result in Table 4 is (65.01 $\pm$ 9.71), without an explanation for the discrepancy. In addition, Table 5 is labeled as VQA but the paper does not

fully describe its training setup, split, or main-task performance. Please identify the dataset, seeds, checkpoint selection, and evaluation split for each table, and either document Table 5 as an additional VQA 2.0 experiment or correct its labeling. This should largely require clarifying existing logs and protocols.

Minor Weaknesses:
None.

Preliminary Recommendation: 3: Borderline Reject

Preliminary Justification:
The paper asks a well-motivated question and presents a simple, potentially useful design. The frozen-encoder CLEVR experiments and accompanying ablations provide meaningful positive evidence, and the candid discussion of the conflicting Flickr30k result is a strength. However, the current evidence does not separate the effect of direction-specific fusion from the effect of increased fusion capacity. The retrieval protocol also needs a more precise definition, particularly regarding whether representations use known paired items, and several tables need clearer experimental provenance. These issues can be addressed through a small capacity-controlled comparison and clearer reporting rather than larger models or expensive additional experiments. I therefore lean borderline reject, while considering the work potentially acceptable after a focused revision.

Resubmission: Yes, the paper is of reasonable quality but requires some revisions before it can be accepted.

Revisions For Resubmission:
Add a parameter-matched CLEVR comparison; fully define both Flickr30k retrieval protocols and ground the main conclusion in pairwise scoring; and clarify the dataset, seed, checkpoint, and split underlying Tables 2, 4, and 5.

Confidence Level: 3 - Moderate Confidence: The reviewer is reasonably knowledgeable about the topic. They understand the paper and OpenReview (about). [Contact \(/contact\)](#)

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